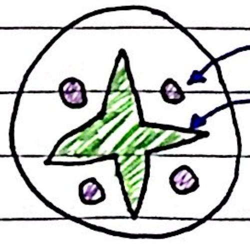
XYLEM AND PHLOEM

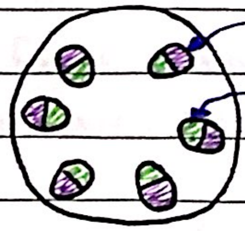
07/11/22

xylem → transport of water and mineral ions, and support.
phloem → transport of sucrose and amino acids

In a root



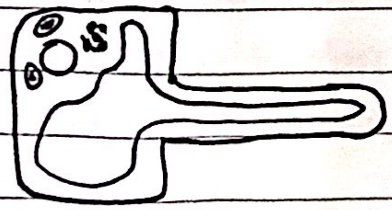
In a stem



structure of xylem:-
- thick walls of lignin → so it can transport the mineral ions
- made of dead cells → no organelles stopping the flow of water
- continous → Long tubelike structure made of singular cells stacked upon each other for easy flow of water.

WATER UPTAKE

11/11/22



* absorbs water & minerals
* stores food
* supports the plant
* Large surface area = more uptake of water & minerals

[pathway of water]:-	[investigating pathway of water]
• root hair cells	• cut a piece of celery
• root cortex	• put in water
• xylem	• add stain (methylene blue or eosin)
• mesophyll cells	• wait few hours - dye will appear in xylem.

TRANSRIRATION

12/11/22

Transpiration

- Loss water vapour from Leaves.
- water evaporates from the surfaces of the mesophyll cells into the air spaces.
- diffuses out the leaves through the stomata

→ water in the mesophyll cells form a thin layer
→ the water evaporates in the air spaces in the spongy mesophyll.
→ this creates a high concentration of water in the air spaces.
→ then the water diffuses out of the leaf, into the surrouding air, through stomata, by diffusion.

* as the water molecules diffuses out of the leaves, more molecules move upward in the xylem as a result of the force of attraction (cohesion) between them.

effect of variation of temperature, wind speed & humidity:-

temperature:-

* warmer = higher rate of transpiration because particles have more energy thus more evaporation

wind speed:-

* more windier = higher rate of transpiration because the diffused particles get carried away faster so the concentration gradient remains high.

humidity:-

* more humidity = less rate of transpiration because more water particles will be

14/11/22

how & why does wilting occur?

- if rate of transpiration of water is more than the rate at which water is being supplied to the plant through absorption, then the plant will start to wilt. Because the cells are not full of water so the don't have the strength to support the plant and so it collapses.

TRANSLOCATION

14/11/22

Translocation

- the movement of sucrose and amino acids in phloem from sources to sinks.

→ sources are the parts of the plant that release sucrose and amino acids.

→ sinks are the parts of the plant that use or store sucrose and amino acids.

explain why some parts of the plant act as a source and sink at different times:-

* when the plant is growing, the source is the roots and the sinks are the many growing areas of the plant.
* when the plant is grown, the leaves are photosynthesizing and producing large quantities of sugars, so the become the source and the roots become the sinks - storing sucrose at starch until it is needed.